

# Buzzard cluster mock data vector with optical-selection effects: forward model, analytical and empirical $\mathcal{B}_{\text{sel}}$ closures

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2026-05-28

## Abstract

This document is the technical reference for `MockDataVector.ipynb`: a Buzzard-halo DES Y3-like cluster mock that layers the Costanzi+2026 (C26) optical selection systematics on top of the raw simulation outputs, on the DES Y1 cluster binning ( $\lambda \in \{20, 30, 45, 60, \infty\}$ ,  $z \in \{0.20, 0.35, 0.50, 0.65\}$ ). The data vector is constructed along *two parallel routes* that share the same Costanzi+2019 (C19) forward model upstream and the same lensing geometry: an **analytical** closure based on the C26 Appendix C Eq. C1 fitting function applied to the C19-binned stack (cheap enough for an MCMC chain), and an **empirical** ratio that stacks on the catalog redMaPPer outputs (`LAMBDA_CHISQ`, `Z_LAMBDA`) and divides by a  $(\log M, z^{\text{tr}})$ -matched random reference (Wu+2022, Costanzi+2026). The notebook also produces a binned number-count vector  $N(\lambda^{\text{ob}}, z^{\text{ob}})$ . Member-galaxy contamination (boost factor) is injected from the published DES Y1 measurements (McClintock+2019), so the saved final shear  $\gamma_t^{\text{mock,obs}} = \Delta\Sigma_{\text{obs,C1}}/B_{\text{Y1}} \cdot \Sigma_{\text{crit}}^{-1}$  mimics the raw uncorrected DES Y1 data vector. All outputs are saved to `output/MockDataVector.npz`; the headline plots embedded in this document live in `output/figs/`.

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## 1 Pipeline topology

The notebook executes the following stages in order:

```
halo_load → photoz_proxy → sample_lambda_true → sample_lambda_obs → lensing_inputs
→ number_counts → analytical_C1 → empirical_mass_match → save_npz
```

The first four stages are the C19 forward model: select host halos above  $\log_{10} M_{\text{vir}} = 13$  in  $z \in [0.20, 0.65]$  (with the Buzzard  $0.33 \leq z < 0.37$  box seam excised), add a DES Y3-like Gaussian photo- $z$  scatter to produce  $z^{\text{ob}}$ , and draw  $\lambda^{\text{tr}}$  and  $\lambda^{\text{ob}}$  from the C26/C19 kernels. `lensing_inputs` centralises the per-halo profiles and  $\Sigma_{\text{crit}}^{-1}$  table, plus the catalog redMaPPer columns (`LAMBDA_CHISQ`, `Z_LAMBDA`) used by the empirical route. The two lensing-data-vector stages are independent and produce their outputs on the same DES Y1 binning so the user can compare them side-by-side. The final stage persists everything to `output/MockDataVector.npz`.

## 2 Shared conventions

**Cosmology.** FlatLambdaCDM( $H_0=70$ ,  $\Omega_m=0.3$ ,  $\Omega_b=0.046$ ,  $T_{\text{cmb}}=2.725$ ) via `astropy.cosmology`, matching the Buzzard simulation cosmology used by the upstream halo catalogue.

**Units.** Per-halo  $\Sigma$ ,  $\Delta\Sigma$  are stored in physical  $M_{\odot}/\text{pc}^2$ ; radii `rp_phys_mpc` (from `src/radial_bins_phys_mpc.py`) are in physical Mpc, with  $N_R = 15$  log-spaced bins between 0.0323 and 30 Mpc. Halo masses  $M_{\text{vir}}$  are in  $h^{-1} M_{\odot}$ .

**Halo selection.** Mirrors 0-MakeMock.ipynb: `pid == -1` (host halos only),  $0 \leq \cos i \leq 1$  (measured), drop the  $0.33 \leq z < 0.37$  simulation-box seam, then require  $0.20 \leq z^{\text{tr}} \leq 0.65$  and  $\log_{10} M_{\text{vir}} \geq 13$ . Yields  $\sim 5.83 \times 10^5$  halos.

**DES Y1 binning (used everywhere).** Both abundance and lensing bins are the DES Y1 cluster edges (Costanzi+2019, McClintock+2019):

$$\text{LBDBINS} = \{20, 30, 45, 60, 500\}, \quad \text{ZMIN\_LIST} = \{0.20, 0.35, 0.50\}, \quad \text{ZMAX\_LIST} = \{0.35, 0.50, 0.65\}.$$

The 500 caps the open  $[60, \infty)$  richness range. There are  $N_{\lambda} \times N_z = 4 \times 3 = 12$  bins.

**Photo- $z$  proxy.** Per-halo Gaussian

$$z^{\text{ob}} = z^{\text{tr}} + \sigma_z(1 + z^{\text{tr}})\mathcal{N}(0, 1), \quad \sigma_z = 0.01, \quad (1)$$

clipped at  $z^{\text{ob}} \geq 0.02$ . Stand-in for the DES Y3 redshift kernel of Myles+2021; captures the dominant scatter but not the tails.

**Random seeds.** A single `numpy.random.default_rng(seed=42)` is shared across the forward model. The MC-vs-PDF sanity histogram uses `seed=123`; the §8.1 bootstrap uses `seed=0`.

### 3 The intrinsic mass–richness relation (C26 Eq. 15)

`Path` `src/costanzi_selection.py` (`l_sat`, `l_tr`, `sample_lambda_true`).

The intrinsic richness  $\lambda^{\text{tr}}$  of a halo (the true number of red-sequence satellite galaxies hosted by the halo, plus the central galaxy) is modelled as a halo central plus a Poisson satellite count. The mean satellite count  $\bar{\lambda}_{\text{sat}}(M, z)$  follows C26 Eq. 15,

$$\bar{\lambda}_{\text{sat}}(M, z) = \left( \frac{M - M_{\text{min}}}{M_{\text{pivot}}} \right)^{\alpha} \left( \frac{1 + z}{1 + z_{\text{piv}}} \right)^{\epsilon}, \quad (2)$$

clipped to zero for  $M < M_{\text{min}}$ . The mean intrinsic richness adds the central galaxy when the halo is above threshold:

$$\langle \lambda^{\text{tr}} | M, z \rangle = \begin{cases} 1 + \bar{\lambda}_{\text{sat}}(M, z), & M \geq M_{\text{min}}, \\ 0, & M < M_{\text{min}}. \end{cases} \quad (3)$$

The DES Y1 NC+3×2pt best-fit constants, hard-coded in `src/costanzi_selection.py`:

$$\log_{10} M_{\text{min}} = 11.385, \quad \alpha = 0.859, \quad \log_{10} M_1 = 12.696, \quad \sigma_{\text{intr}} = 0.181, \quad \epsilon = 0.284, \quad z_{\text{piv}} = 0.4544,$$

with  $M_{\text{pivot}} = M_1 - M_{\text{min}}$ .

The per-halo  $\lambda^{\text{tr}}$  value is drawn (C26 Sec. III) by (i) broadening the Poisson rate  $\bar{\lambda}_{\text{sat}}$  with a multiplicative log-normal of width  $\sigma_{\text{intr}}$ , (ii) drawing an integer satellite count, and (iii) adding 1 for the central galaxy if the halo is above  $M_{\text{min}}$ :

$$\tilde{\lambda}_{\text{sat}} = \bar{\lambda}_{\text{sat}}(M, z) \exp\left[\sigma_{\text{intr}} \mathcal{N}(0, 1) - \frac{1}{2}\sigma_{\text{intr}}^2\right], \quad N_{\text{sat}} \sim \text{Poisson}(\tilde{\lambda}_{\text{sat}}), \quad \lambda^{\text{tr}} = N_{\text{cen}}(M) + N_{\text{sat}},$$

where  $N_{\text{cen}}(M) = 1$  if  $M \geq M_{\text{min}}$  and 0 otherwise.

### 4 The projection kernel (C19 Eq. 6)

`Path` `src/costanzi_selection.py` (`projection_params`, `sample_lambda_obs`). C19 Eq. 6 is a Gaussian-core + exponentially-modified-Gaussian (EMG) mixture:

$$P(\lambda^{\text{ob}} | \lambda^{\text{tr}}, z) = (1 - f_{\text{prj}}) \mathcal{N}(\mu, \sigma) + f_{\text{prj}} \text{EMG}(\mu, \sigma, \tau). \quad (4)$$

The four parameters  $\{\mu, \sigma, \tau, f_{\text{prj}}\}$  depend on  $\lambda^{\text{tr}}$  with  $z$ -dependent coefficients calibrated from SDSS redMaPPer (C19 Appendix A, Eq. A8):

$$\begin{aligned} \mu(\lambda^{\text{tr}}, z) &= a_{\mu}(z) + b_{\mu}(z) \lambda^{\text{tr}}, & \sigma(\lambda^{\text{tr}}, z) &= b_{\sigma}(z) (\lambda^{\text{tr}})^{a_{\sigma}(z)}, \\ \tau(\lambda^{\text{tr}}, z) &= b_{\tau}(z) (\lambda^{\text{tr}})^{-a_{\tau}(z)}, & f_{\text{prj}}(\lambda^{\text{tr}}, z) &= b_{f_{\text{prj}}}(z) (1 + e^{-\lambda^{\text{tr}}})^{-a_{f_{\text{prj}}}(z)}. \end{aligned} \quad (5)$$

The 15 rows of `data/prj_params_DESY3_lss_lin_dep_getdist_v1.txt` are interpreted as posterior samples and collapsed to the posterior mean via `load_prj_posterior_mean`.

The mixture is sampled directly using the  $\text{EMG} = \mathcal{N}(\mu, \sigma) + \text{Exp}(1/\tau)$  representation, avoiding a closed-form EMG density evaluation in the sampling path. The closed form is used only for the sanity overlay below.

**Sanity check 1 — MC draws vs analytical kernel.** Histogram repeated `sample_lambda_obs` draws at fixed  $(\lambda^{\text{tr}}, z)$  and overlay the closed-form C19 PDF; they should agree within Poisson noise.

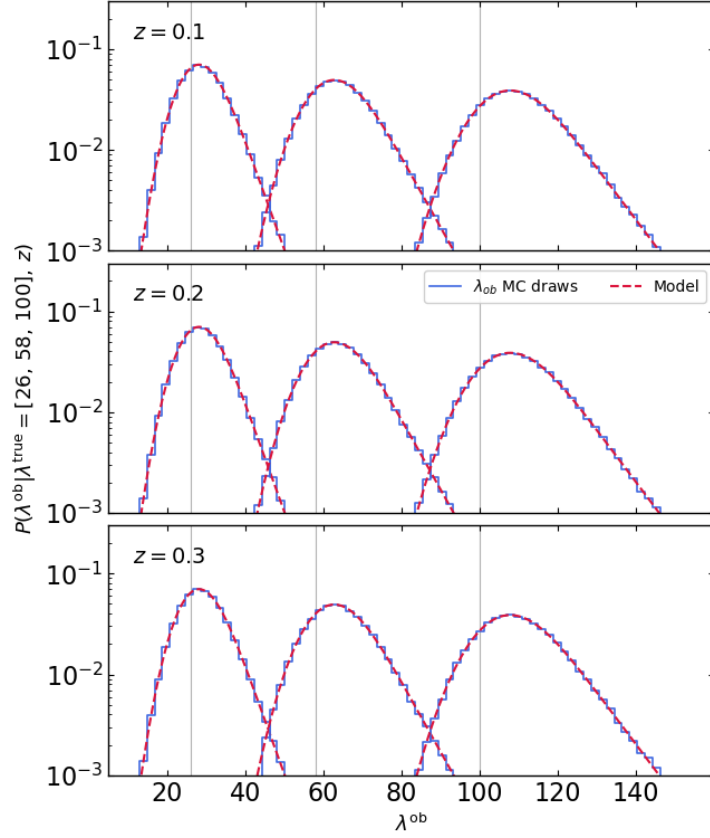


Figure 1: MC draws of  $\lambda^{\text{ob}}$  from `sample_lambda_obs` (blue histograms) overlaid with the closed-form PDF of Eq. 4 (red dashed) at three reference redshifts and three intrinsic richnesses. The two agree within Poisson noise across the bulk and the tail.

**Sanity check 2 — Gaussian core + EMG tail decomposition.** Same kernel as Eq. 4, split into the two physical components. At large  $\lambda^{\text{tr}}$  the EMG tail dominates the high- $\lambda^{\text{ob}}$  side and drives the projection bias; this is what the analytical  $\mathcal{B}_{\text{sel}, \text{C1}}$  of §6 fits.

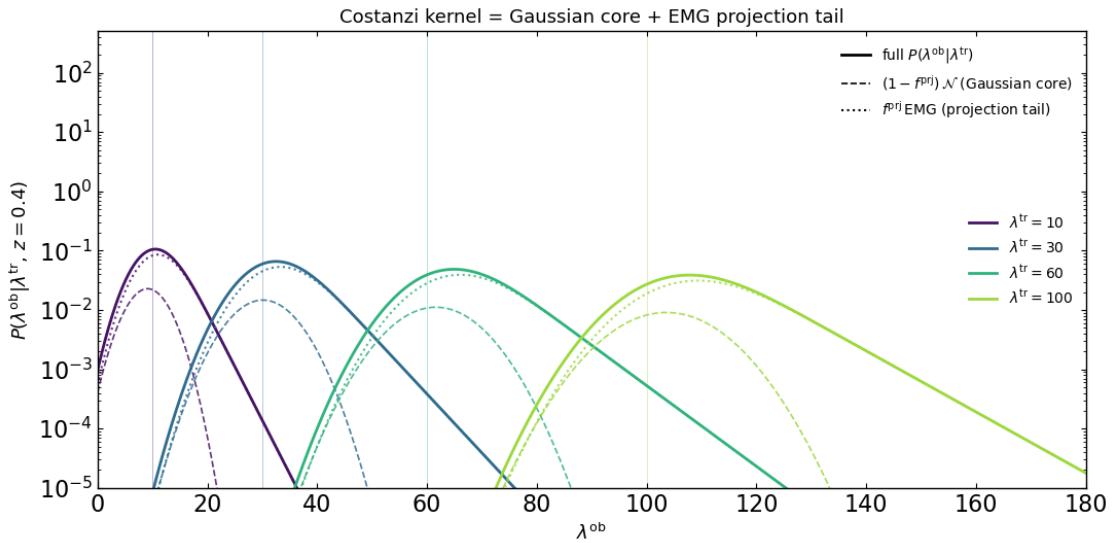


Figure 2: Decomposition of the kernel of Eq. 4 into its  $(1 - f_{\text{prj}})\mathcal{N}$  Gaussian core (dashed) and  $f_{\text{prj}}$  EMG projection tail (dotted) at  $z = 0.4$  for four intrinsic richnesses  $\lambda^{\text{tr}} \in \{10, 30, 60, 100\}$ . The EMG tail rises with  $\lambda^{\text{tr}}$  and drives the optical-selection bias on the lensing data vector.

## 5 Output 1 — $N(\lambda^{\text{ob}}, z^{\text{ob}})$ number counts

We obtain  $N(\lambda^{\text{ob}}, z^{\text{ob}})$  by counting the mock halos that fall into each  $(\lambda^{\text{ob}}, z^{\text{ob}})$  bin on the DES Y1 grid. Both axes are forward-model outputs rather than catalog redMaPPer columns: the C19 sampler depends only on  $(M, z)$  and carries no per-halo LSS information, which matches the inputs of the analytical  $\mathcal{B}_{\text{sel}, \text{C1}}(\lambda^{\text{ob}}, z^{\text{ob}})$  of §6 and keeps the abundance and lensing sides of the data vector in the same C26/C19 model world.  $N(\lambda^{\text{ob}}, z^{\text{ob}})$  is therefore the correct entry for a C26 likelihood integrating Eq. 2 into Eq. 4.

## 6 Lensing data vector — ANALYTICAL route (Matteo / C26 Eq. C1)

**Binning quantity.** The C19 forward-model  $(\lambda^{\text{ob}}, z^{\text{ob}})$ . Each  $\lambda^{\text{ob}}$  is an independent draw from the kernel of Eq. 4 conditioned only on  $(\lambda^{\text{tr}}, z)$ : the kernel encodes the LSS-driven projection statistics through its EMG tail at the population level, but per-halo  $\lambda^{\text{ob}}$  values are not correlated with the actual line-of-sight structure around each halo and therefore carry no information correlated with the per-halo lensing signal.

**Selection bias.** redMaPPer counts red-sequence galaxies inside a projected aperture with no line-of-sight resolution, so projected LSS preferentially boosts  $\lambda^{\text{RM}}$  for halos in over-dense filaments and lifts the stacked  $\Delta\Sigma(R)$  above the mass-only expectation. C26 Appendix C captures this lift with a 5-parameter fitting function (no integration); Matteo (priv. comm.) reports the same form fits the bias on  $\Delta\Sigma$ :

$$\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R) = A \left( \frac{R}{R_0} \right)^\alpha \left[ 1 + \left( \frac{R}{R_0} \right)^\gamma \right]^{(\beta-\alpha)/\gamma} + C, \quad (6)$$

with  $(A, \alpha, \beta, \gamma, C) = (0.12, 4.11, -0.18, 1.82, 1.00)$ ,  $R_0 = R_\lambda(\lambda^{\text{ob}})(1+z^{\text{ob}})$ , and  $R_\lambda = (\lambda^{\text{ob}}/100)^{0.2} h^{-1} \text{ Mpc}$  (C26 Appendix B).

The C26 paper writes the published form for  $\Sigma$  with a fixed +1 rather than a free  $C$  ( $C = 1$  limit of Eq. 6); we plot both flavours for comparison. The data vector is

$$\Delta\Sigma_{\text{obs}}(R) = \mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R) \Delta\Sigma_{\text{stack}}(R), \quad \langle \gamma_t(R) \rangle = \Delta\Sigma_{\text{obs}}(R) \Sigma_{\text{crit}}^{-1} \left( \langle z^{\text{ob}} \rangle_{\text{bin}} \right), \quad (7)$$

with  $\Delta\Sigma_{\text{stack}}$  the unweighted mean of `DeltaSigma` across the C19-binned halos in each  $(\lambda^{\text{ob}}, z^{\text{ob}})$  cell.

**Why a fitting function?** An MCMC chain evaluates the lensing model  $\sim 10^4$ – $10^6$  times, so the per-step cost matters. Computing the full Costanzi-2026 prediction from scratch requires nested integrals over halo mass and redshift at every step — prohibitive at chain length. Equation (6) replaces those integrals with five constants per bin, cutting the cost from seconds to microseconds without changing the answer.

**Selection-bias ratio  $\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R)$**  The fitting function evaluated at the bin-mean  $(\langle \lambda^{\text{ob}} \rangle, \langle z^{\text{ob}} \rangle)$  for every  $(\lambda^{\text{ob}}, z^{\text{ob}})$  cell, with the published  $\Sigma$ -flavour overlay.

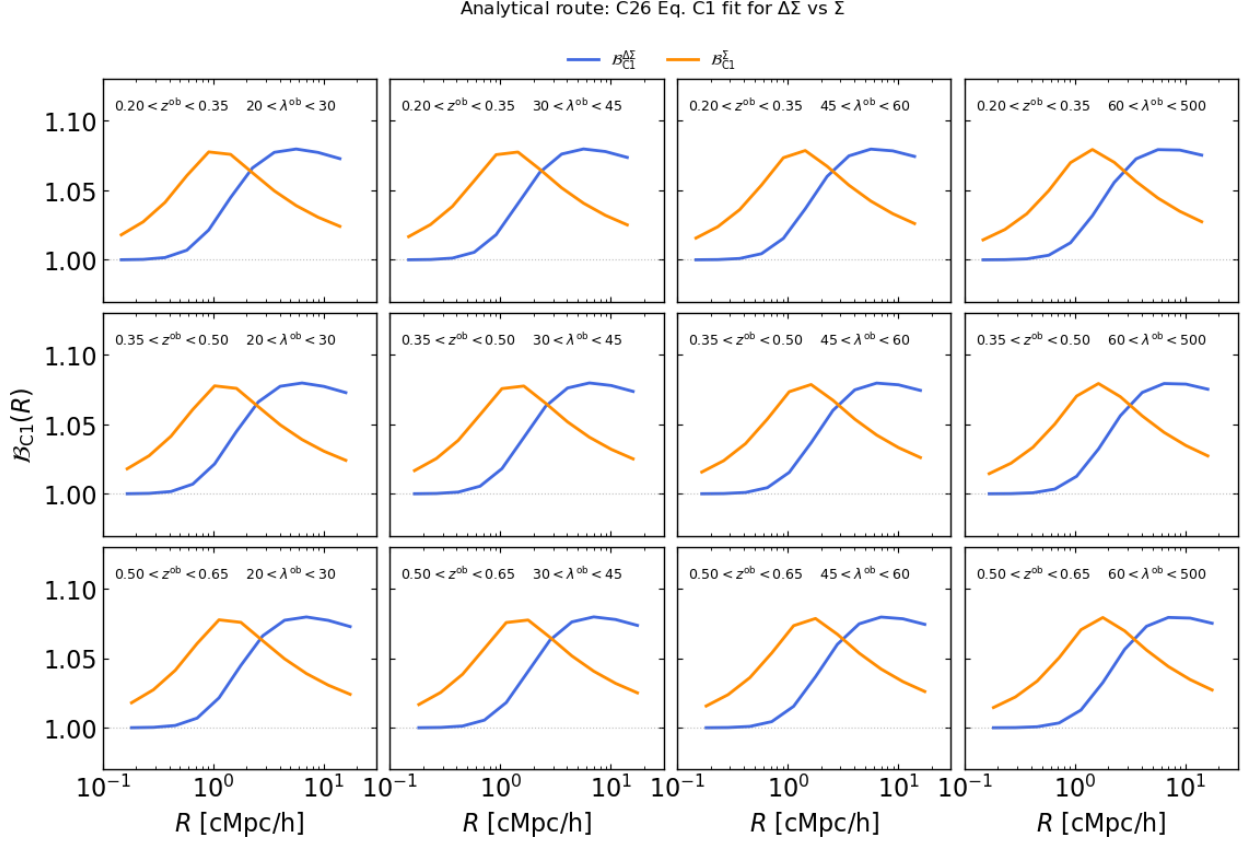


Figure 3: Analytical route: C26 Eq. C1 selection-bias ratio per  $(\lambda^{\text{ob}}, z^{\text{ob}})$  bin on the DES Y1 grid. Blue:  $\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}$  (used in the data vector). Orange:  $\mathcal{B}_{\text{sel}, \text{C1}}^{\Sigma}$  overlay.  $x$ -axis is comoving  $R$  in  $h^{-1}$  Mpc to match the C26 convention;  $y$ -axis is the bias factor.

### 6.1 Boost factor: member-galaxy contamination $B(R)$

A real DES Y1 measurement of  $\Delta\Sigma$  is diluted by cluster-member galaxies that survive photo- $z$  cuts and end up in the lensing source catalog. These members carry no shear signal and reduce the measured profile by the boost factor  $B(R) \equiv 1/(1 - f_{\text{cl}}) \geq 1$  (McClintock+2019 §5.3.1).

$$B(R; r_s, b_0) = 1 + b_0 \frac{1 - f(x)}{x^2 - 1}, \quad x = R/r_s,$$

$$f(x) = \begin{cases} \frac{\arctan \sqrt{x^2 - 1}}{\sqrt{x^2 - 1}}, & x > 1 \\ \frac{\operatorname{arctanh} \sqrt{1 - x^2}}{\sqrt{1 - x^2}}, & x < 1 \end{cases} \quad (8)$$

To make the saved data vector mock the *raw uncorrected* DES Y1 measurement, both lensing routes have their final shear stack divided by  $B(R)$ :

$$\langle \gamma_t \rangle^{\text{mock, obs}} = \frac{\langle \gamma_t \rangle^{\text{obs}}}{B_{\text{Y1 data}}(R)} \quad (9)$$

**Where  $B_{\text{Y1 data}}(R)$  comes from.** The DES Y1 cluster analysis released measured boost-factor profiles per richness/redshift bin, located at `data/boost_factor/profiles/ (full-unblind-v2-mcal-zmix.y1clust_1{1}-z{`. The files are tabulated on an 11-point physical-Mpc  $R$  grid (0.166 to 15.80 Mpc); we adopt this grid as the global radial axis for the saved data vector. The 7 cosmology-side richness bins 10..16 include

$\lambda < 20$  ranges; only 13..16 match our 4 lensing bins  $\lambda \in \{[20, 30), [30, 45), [45, 60), [60, \infty)\}$ . The parametric McClintock+2019 Table B1 fit is also computed (cell 42) for sanity but is not used in the saved output.

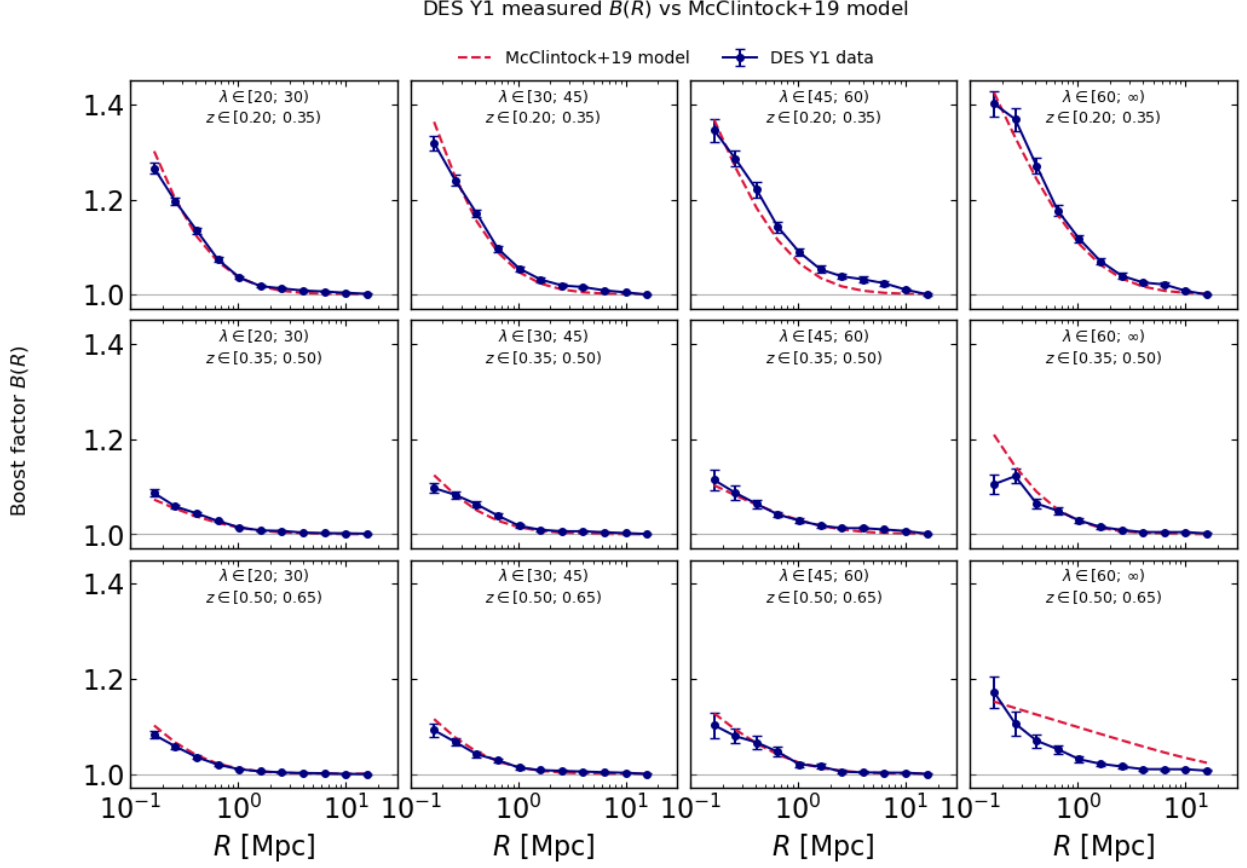


Figure 4: Measured DES Y1 boost factor  $B(R)$  per  $(\lambda^{\text{RM}}, z^{\text{RM}})$  bin with reported errors. Small- $R$  amplitude rises with richness and falls with redshift, as expected for a red-sequence-driven member contamination.

## 7 Lensing data vector — EMPIRICAL route (Heidi / mass-matched ratio)

**Binning quantity.** The catalog redMaPPer outputs  $(\lambda^{\text{RM}}, z^{\text{RM}})$ , i.e. `LAMBDA_CHISQ` and `Z_LAMBDA`. These come from running the redMaPPer cluster finder on the Buzzard photometric catalog and *do* carry per-halo LSS information.

**Selection bias.** Built directly from the catalog as the ratio of the redMaPPer-selected stack to a  $(\log M, z^{\text{tr}})$ -matched random reference, following Wu+2022 method iii (`src/stacked_profile_weighted_by_mass_redsh`). For each  $(\lambda^{\text{RM}}, z^{\text{RM}})$  bin:

$$\langle \Delta \Sigma(R) \rangle_{\text{rnd}} = \frac{\sum_c w_c \langle \Delta \Sigma(R) | c \rangle_{\text{full mock}}}{\sum_c w_c}, \quad w_c = N_c^{\text{sel}}, \quad (10)$$

summed over  $(\log_{10} M, z^{\text{tr}})$  cells of width  $(\Delta \log M, \Delta z) = (0.1, 0.05)$ . The cell averages are taken over the *full* mock, so the reference resamples halos that share the selected sample’s mass and redshift but



were never preferentially picked for their environment. The empirical bias is

$$\mathcal{B}_{\text{sel}}^{\text{emp}}(R) = \frac{\langle \Delta\Sigma(R) \rangle_{\text{sel}}}{\langle \Delta\Sigma(R) \rangle_{\text{rnd}}}.$$

No fitting function, no  $R \rightarrow 0$  extrapolation, both numerator and denominator come straight from DeltaSigma.

**Bootstrap.** The reported  $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$  is the **bootstrap mean** over  $N_{\text{boot}} = 50$  resamples of halos within each bin (with mass-matched reference rebuilt at every step), capturing the statistical uncertainty driven by finite halo count. The shaded bands in the panel grids are the bootstrap standard deviation; the saved file keeps only the bootstrap mean.

### 7.1 Selection-bias ratio $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$ with bootstrap bands

We evaluate  $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$  on the redMaPPer-binned  $(\lambda^{\text{RM}}, z^{\text{RM}})$  grid using Eq. (10) and report the bootstrap mean and  $1\sigma$  band per bin. The expected signature is the bell-shaped excess peaking near  $R \sim 1$  pMpc that Wu+2022 (Fig. 2) identify as the LSS-environment-driven selection bias: at smaller scales the one-halo term dominates and the ratio approaches unity, while at larger scales the projected over-density that boosted  $\lambda^{\text{RM}}$  falls outside the lensing aperture and the ratio relaxes back toward one.

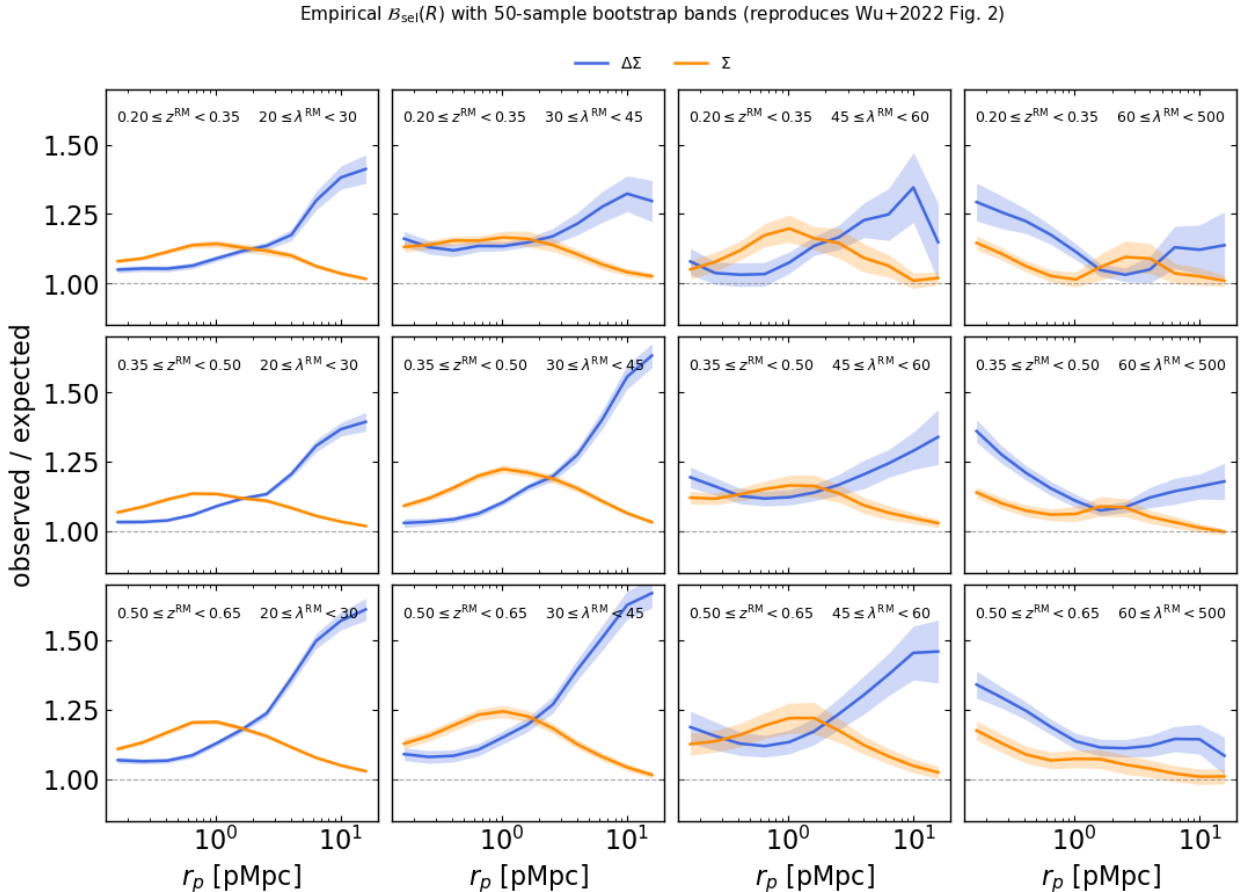


Figure 5: Empirical route:  $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$  per  $(\lambda^{\text{RM}}, z^{\text{RM}})$  bin on the DES Y1 grid, with  $N_{\text{boot}} = 50$  bootstrap bands. Blue:  $\Delta\Sigma$  ratio. Orange:  $\Sigma$  ratio (overlay). Reproduces Wu+2022 Fig. 2: the bell-shaped peak around  $R \sim 1$  pMpc is the LSS-environment-driven selection bias signal.



## 7.2 Null test: replace $\lambda^{\text{RM}}$ with the C19 sampler

If we re-bin on the C19 forward-model ( $\lambda^{\text{ob}}, z^{\text{ob}}$ ) instead of the catalog ( $\lambda^{\text{RM}}, z^{\text{RM}}$ ), the same mass-matched-random pipeline should return unity at every  $R$ , because the C19 sampler is by construction blind to per-halo LSS. Any residual deviation from 1 is the bias of the matching procedure itself, not a physical signal.

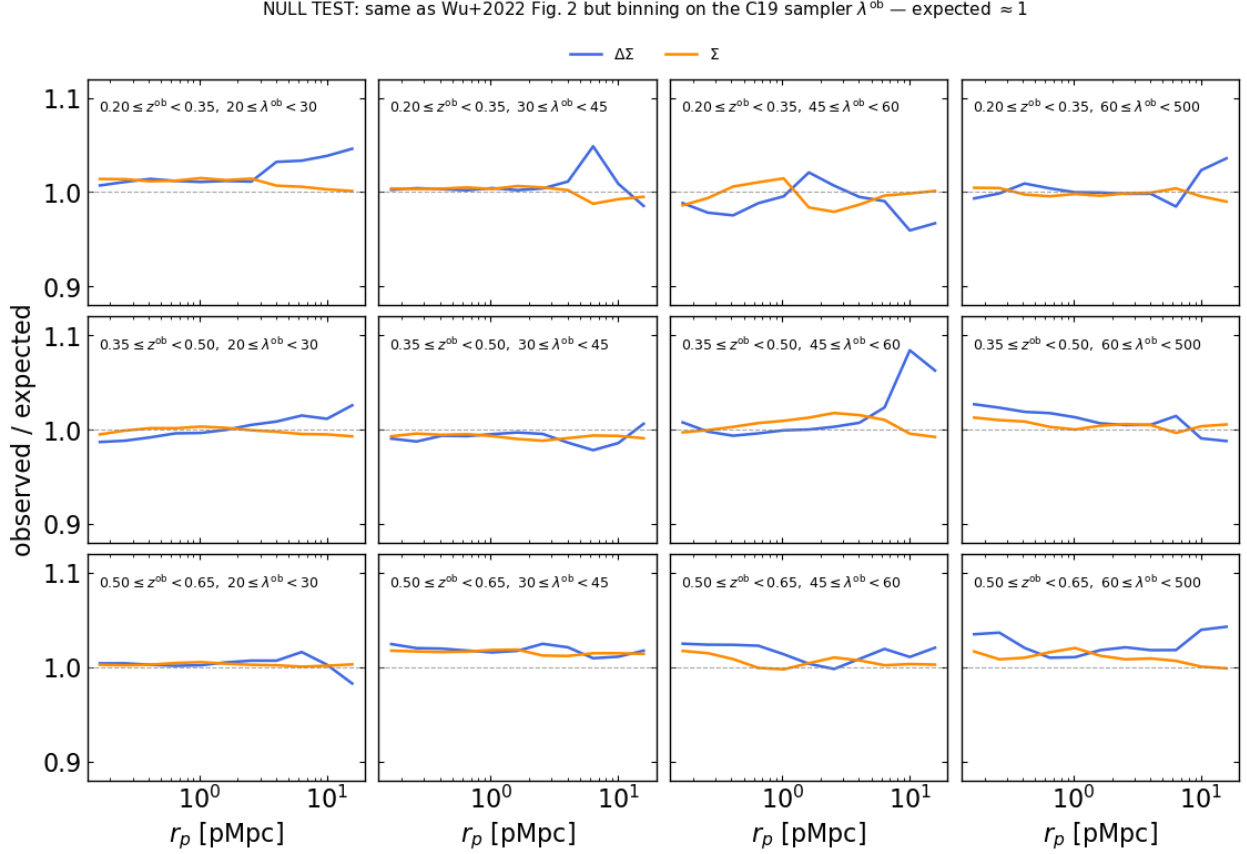


Figure 6: Null test: same Wu+22 panel grid but binning on the C19 sampler ( $\lambda^{\text{ob}}, z^{\text{ob}}$ ) rather than the catalog ( $\lambda^{\text{RM}}, z^{\text{RM}}$ ). The bell collapses to  $\approx 1$  across all bins, confirming the  $R \sim 1$  pMpc peak in Fig. 5 is genuinely LSS-driven.

## 7.3 Tangential shear: redMaPPer-selected vs mass-matched random

The bias ratio of §7.1 is most directly seen by comparing the two tangential-shear stacks that enter its numerator and denominator. The mass-matched random reference is the  $\gamma_t$  a hypothetical unbiased selector would measure on halos of the same  $(\log M, z^{\text{tr}})$  distribution; the redMaPPer-selected stack is the  $\gamma_t$  an actual cluster finder produces. Their offset is the observable counterpart of  $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$  and the quantity an MCMC must reproduce.

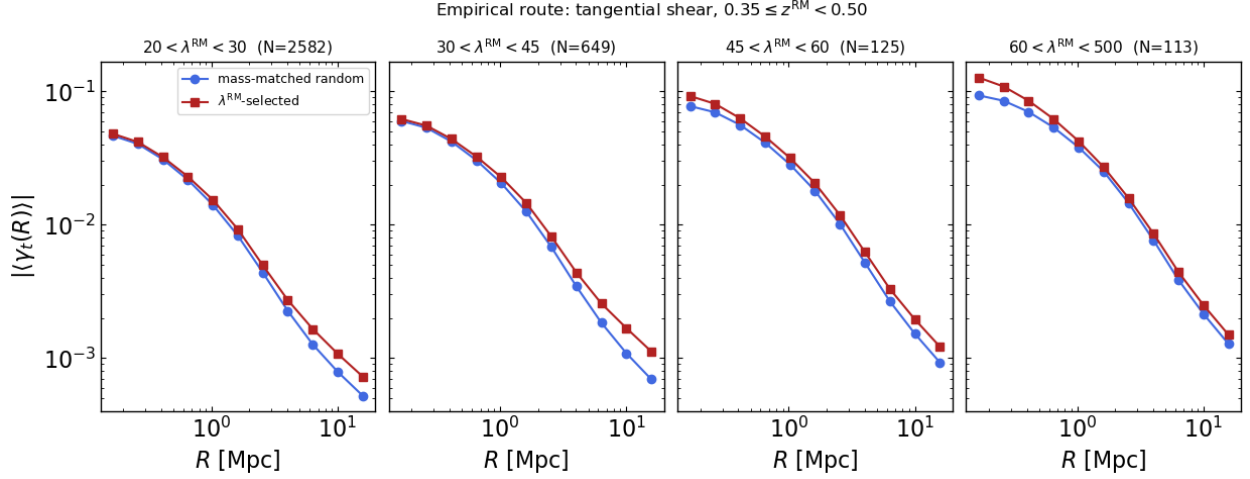


Figure 7: Empirical route: per-bin tangential shear in the middle redshift slice  $z^{\text{RM}} \in [0.35, 0.50)$ . Blue circles:  $\gamma_t$  from the mass-matched random reference. Red squares:  $\gamma_t$  from the  $\lambda^{\text{RM}}$ -selected stack (the observed signal). The selected curve sits above the reference by the  $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$  factor of Fig. 5.

## 8 Output archive

The final notebook cell writes both routes plus bin metadata to a single `output/MockDataVector.npz`. With  $N_\lambda = 4$ ,  $N_z = 3$ ,  $N_R = 11$  (DES Y1 boost-factor profile grid):

| Key                                                            | Shape                        | Content                                                                                      |
|----------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------|
| <i>Bin metadata</i>                                            |                              |                                                                                              |
| <code>radii_phys_mpc</code>                                    | $(N_R,)$                     | 11-pt DES Y1 boost grid (physical Mpc; 0.166–15.80)                                          |
| <code>lambda_bins</code>                                       | $(N_\lambda + 1,)$           | DES Y1 richness edges $\{20, 30, 45, 60, 500\}$                                              |
| <code>z_bin_min</code>                                         | $(N_z,)$                     | lower-redshift edges                                                                         |
| <code>z_bin_max</code>                                         | $(N_z,)$                     | upper-redshift edges                                                                         |
| <i>Number counts (DES Y1 expectation)</i>                      |                              |                                                                                              |
| <code>NC</code>                                                | $(N_\lambda, N_z)$           | $N(\lambda^{\text{ob}}, z^{\text{ob}})$ , Buzzard counts rescaled by $\Omega_{\text{Y1}}(z)$ |
| <i>Analytical route, C19-binned</i>                            |                              |                                                                                              |
| <code>gamma_t_stack_C19</code>                                 | $(N_\lambda, N_z, N_R)$      | stacked $\gamma_t$ , no correction                                                           |
| <code>gamma_t_obs_C1</code>                                    | $(N_\lambda, N_z, N_R)$      | stack $\times \mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}$                            |
| <code>B_sel_C1</code>                                          | $(N_\lambda, N_z, N_R)$      | analytical $\mathcal{B}_{\text{sel}, \text{C1}}^{\Delta\Sigma}(R)$                           |
| <i>Empirical route, RM-binned</i>                              |                              |                                                                                              |
| <code>gamma_t_stack_RM</code>                                  | $(N_\lambda, N_z, N_R)$      | $(\log M, z^{\text{tr}})$ -matched random reference $\gamma_t$                               |
| <code>gamma_t_obs_RM</code>                                    | $(N_\lambda, N_z, N_R)$      | $\lambda^{\text{RM}}$ -selected stack $\gamma_t$ (the observed signal)                       |
| <code>B_sel_emp</code>                                         | $(N_\lambda, N_z, N_R)$      | bootstrap-mean $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$                                    |
| <i>DES Y1 boost factor + final mock-observed shears (§6.1)</i> |                              |                                                                                              |
| <code>B_data</code>                                            | $(N_\lambda, N_z, N_R)$      | measured DES Y1 $B(R)$                                                                       |
| <code>B_data_err</code>                                        | $(N_\lambda, N_z, N_R)$      | 1- $\sigma$ error per radial bin                                                             |
| <code>B_data_cov</code>                                        | $(N_\lambda, N_z, N_R, N_R)$ | full radial covariance matrix per bin                                                        |
| <code>gamma_t_mock_obs_C1</code>                               | $(N_\lambda, N_z, N_R)$      | analytical (Matteo) shear with Y1 boost dilution                                             |
| <code>gamma_t_mock_obs_RM</code>                               | $(N_\lambda, N_z, N_R)$      | empirical (Heidi) shear with Y1 boost dilution                                               |

Table 1: Output-archive layout. Default shapes are  $(4, 3)$  for the counts and  $(4, 3, 15)$  for the radial vectors.

## 9 Known simplifications

- **Photo- $z$  kernel.** The Gaussian  $\sigma_z = 0.01 (1 + z^{\text{tr}})$  proxy is a stand-in for the redshift- and colour-dependent kernel of Myles+2021. It captures the dominant scatter but misses the tails that drive the high- $z$  outlier rate.
- **Boost factor uses DES Y1, not Y3, measurements.** §6.1 injects  $B(R)$  from McClintock+2019 Y1 profiles, which is the closest released measurement matching our cluster binning. A Y3 boost-factor measurement is not yet released; when it is, swap the source files in `data/boost_factor/profiles/` without further code change.
- **Posterior-mean projection coefficients.** The C19 projection parameters are evaluated at the posterior mean of the upstream chain. Row-by-row covariance propagation is not implemented.
- **Single Y3 realisation.** The §8.1 empirical ratio is computed on one Y3 realisation; Wu+2022 averages 12 Y1 + 1 Y3 realisations and is correspondingly tighter.

## 10 File map

- `MockDataVector.ipynb` — the production notebook. Top-to-bottom execution reproduces every figure and writes the output archive.
- `src/costanzi_selection.py` — C26 Eq. 15 sampler and C19 Eq. 6 mixture sampler, plus the legacy single-scale  $\mathcal{B}_{\text{sel}}$  weight.
- `src/stacked_profile_weighted_by_mass_redshift.py` — Wu+22 / Esteves ( $M, z$ )-matched random reference builder.
- `src/radial_bins_phys_mpc.py` — the 15-bin physical radial grid `rp_phys_mpc`.
- `data/prj_params_DESY3_lss_lin_dep_getdist_v1.txt` — 15 posterior samples of the 10 C19 projection coefficients.
- `src/fileLoc.py` — machine-resolved path table for the raw Buzzard halo catalogue, profile output, and  $\beta_{\text{eff}}$  table.
- `output/MockDataVector.npz` — saved data vector (Tab. 1).
- `output/figs/` — headline plots embedded in this PDF.
- `docs/MockDataVector.tex` — this document.

## 11 References

- Costanzi et al. 2026, *Forward analytical model for the optical selection bias on galaxy cluster lensing profiles*. Eqs. 15 (mass–richness), Appendix C Eq. C1 (selection-bias fit), Sec. III (Poisson–Gaussian convolution).
- Costanzi et al. 2019, arXiv:1807.07072. Eq. 6 (Gaussian + EMG mixture), Appendix A Eq. A8 (per-parameter forms).
- Wu et al. 2022, arXiv:2203.05416. Fig. 2 (selection-bias panel grid) and Appendix B (mass–redshift weighted reference).
- McClintock et al. 2019, MNRAS 482, 1352 — DES Y1 cluster binning convention used throughout.